Thermodynamic description of the Al–Ni system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Ni binary system as calculated with pycalphad from the SGTE dataset AlNi-14Zhu.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L + (Ni) → Ni3Al, lies at 1643.1 K.

1. Introduction

The Al–Ni system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, order disorder, congruent compound. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of J. Zhu, M.S. Titus, T.M. Pollock, J. Phase Equilib. Diffus., 35, 595-611(2014). The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Al reproduced by the dataset is 939.4 K. The melting point of Ni reproduced by the dataset is 1728.2 K.

2. Phase diagram

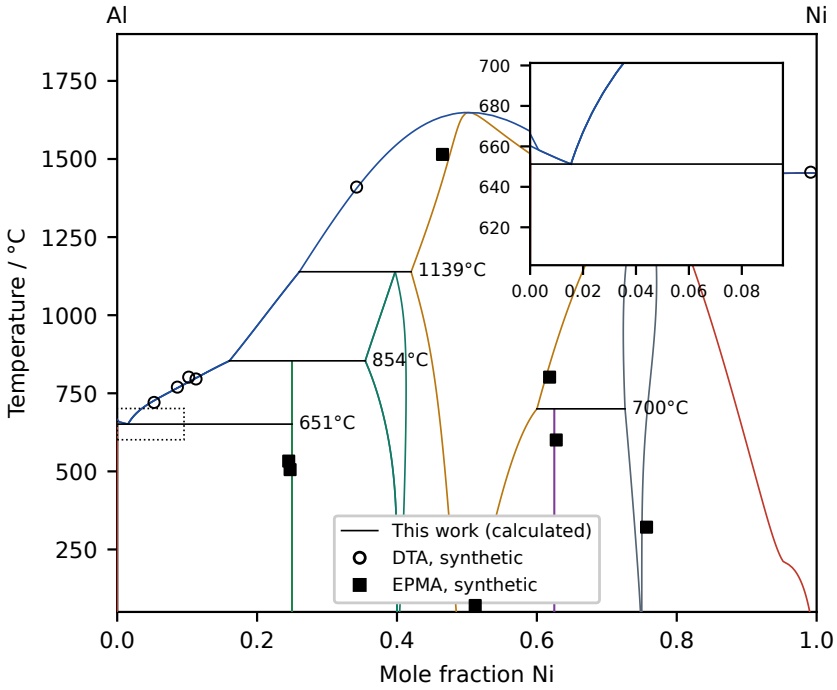


Fig. 1. Calculated Al–Ni phase diagram with synthetic experimental-style points (inset: enlarged view near the lowest invariant).

The dataset places a peritectic (L + (Ni) → Ni3Al) at 1643.1 K; the composition of L is x(Ni) = 0.759. Eutectic equilibrium L → NiAl + Ni3Al is found at 1642.3 K and x(Ni) = 0.744 (L). The dataset places a peritectic (L + NiAl → Al3Ni2) at 1412.2 K; the composition of L is a mole fraction of 0.261. The peritectic reaction L + Al3Ni2

→ Al3Ni is calculated at 1127.0 K, with L at x(Ni) = 0.160. Peritectoid equilibrium NiAl + Ni3Al → Al3Ni5 is found at 973.4 K and x(Ni) = 0.625 (Al3Ni5). We obtain 924.4 K for the eutectic reaction (L → (Al) + Al3Ni), where L has a mole fraction of 0.016. Ni3Al melts congruently at 1643.1 K and x(Ni) = 0.761.

Table 1. Invariant reactions in the Al–Ni system (calculated).

Reaction	Type	T (°C)	x(Ni), 1	x(Ni), 2	x(Ni), 3
L + (Ni) → Ni3Al	peritectic	1369.9	0.759	0.786	0.761
L → NiAl + Ni3Al	eutectic	1369.2	0.744	0.712	0.746
L + NiAl → Al3Ni2	peritectic	1139.1	0.261	0.420	0.397
L + Al3Ni2 → Al3Ni	peritectic	853.9	0.160	0.355	0.250
NiAl + Ni3Al → Al3Ni5	peritectoid	700.2	0.600	0.726	0.625
L → (Al) + Al3Ni	eutectic	651.2	0.016	0.000	0.250

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 2000 K the integral mixing enthalpy of the liquid has its minimum, −49.27 kJ/mol, at x(Ni) = 0.500. The calculated enthalpy of formation of NiAl at 298.15 K is −62.24 kJ/mol per mole of atoms.

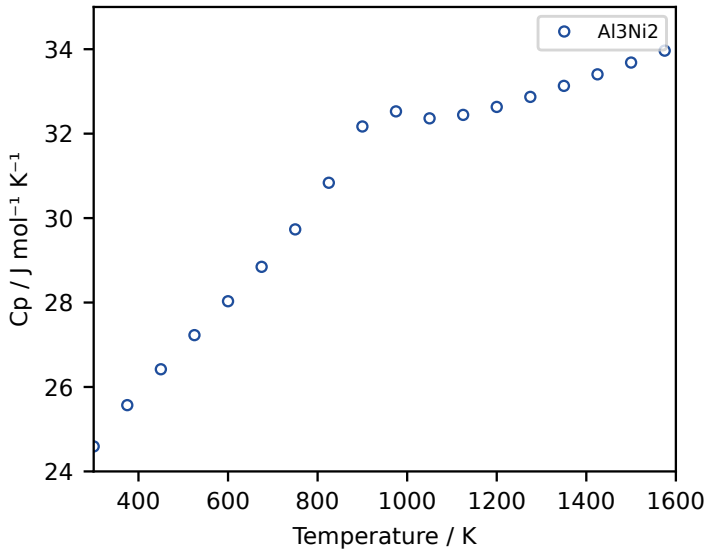


Fig. 2. Calculated heat capacity of Al3Ni2.

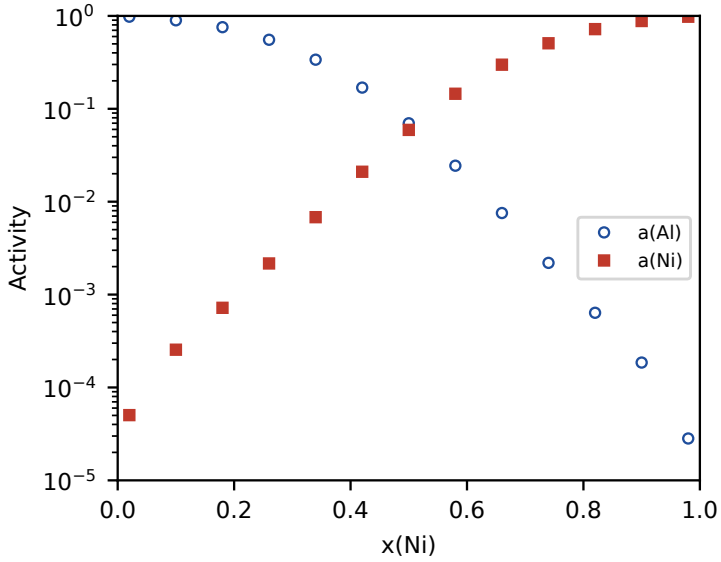


Fig. 3. Calculated activities in the liquid at 2000 K (reference: pure liquids).

Table 2. Calculated formation properties of the intermediate phases (Al–Ni).

Phase	at.% Ni	ΔfH(298 K) / kJ mol⁻¹	S(298 K) / J mol⁻¹ K⁻¹
Al3Ni2	39.5	-60.28	21.63
Al3Ni5	62.5	-53.39	25.96
Al3Ni	25.0	-43.45	21.14
NiAl	53.4	-62.24	24.73
Ni3Al	86.6	-21.39	28.80

Reference: stable elements (SER) at 298.15 K.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.